

Robots That 'Grow Up'

Programming devices—and robots—to do stuff is no longer exciting. The next thing on the wish list is robots that learn how to behave and do things autonomously, just like humans and animals learn as they grow up

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Shape-shifting robots, 3D printed ones, those with facial recognition and social skills too are evolving in robotics labs across the world, but there is something more interesting brewing! Scientists are now working to develop robots that can learn to do things and react to situations like humans do. Just as a baby learns to be and do, by observing, assimilating, experimenting and adapting, robots of the future can be taught, or will even learn naturally like we do, without being programmed to do everything. Here, we look at some research and early examples.

Hi Jimmy!

You need to 3D print some parts, fix in other non-printable parts including an Intel Edison processor, and voila, your robot buddy is ready! And, according to Intel, you do not need a PhD to program Jimmy, their open source robot, which will be on sale by end of this year.

When first launched, Jimmy walked up on stage, introduced himself, danced and then sat down to watch the rest of the proceedings. A short trip to Intel's website and you can see Jimmy solving the Rubik's cube, cheering for his favourite team at a match, and making a public appearance with Intel CEO. Once you make your own

Jimmy (or whatever you call it), it can send tweets on your behalf, sing or dance with you, bring you food, and more.

Led by Brian David Johnson, Intel's robotics lab ultimately aims to build an easy-to-make and easy-to-use, completely open source robot that is extremely social, friendly, and able to dream, feel and think. While it seems awesome and impossible that you can make your own robot and also make it do whatever you want, Johnson

explains that it is just akin to a smartphone with customisable apps.

The company is also betting big on the open source model, because when you have a smart and social robot like Jimmy, it needs to constantly evolve, and what better way to do it than have the whole world contribute apps and ideas?

Please, Monica...

Do you remember walking into your dad's office, to be greeted by his personal assistant, who politely seated you, offered your favourite beverage and book, chatted friendly and then subtly broke the news that your father was out of office for a meeting? Do you remember her trying to step in and solve small problems when dad was overworked, booking tickets, paying bills and helping out with professional and personal tasks that could be shared? Where did they go? Why aren't today's smartphones and laptops as efficient as those reliable personal secretaries in reducing your stress? Well, perhaps it is because these inanimate objects cannot think, feel or even actually realise how tired or stressed you are!

You would not feel this way if you had a couple of digital assistants like Eric Horvitz's. Horvitz, managing director of Microsoft Research Redmond, is deeply involved in the Situational Interaction project to enable



Intel's vision of tomorrow's robot—Jimmy (Courtesy: Intel)